

Task 15: Vulnerability Assessment & Risk Prioritization

Tool Used

Primary Tool: Nessus Essentials

Operating System Used: Ubuntu Linux

1. Define Scope and Target Systems

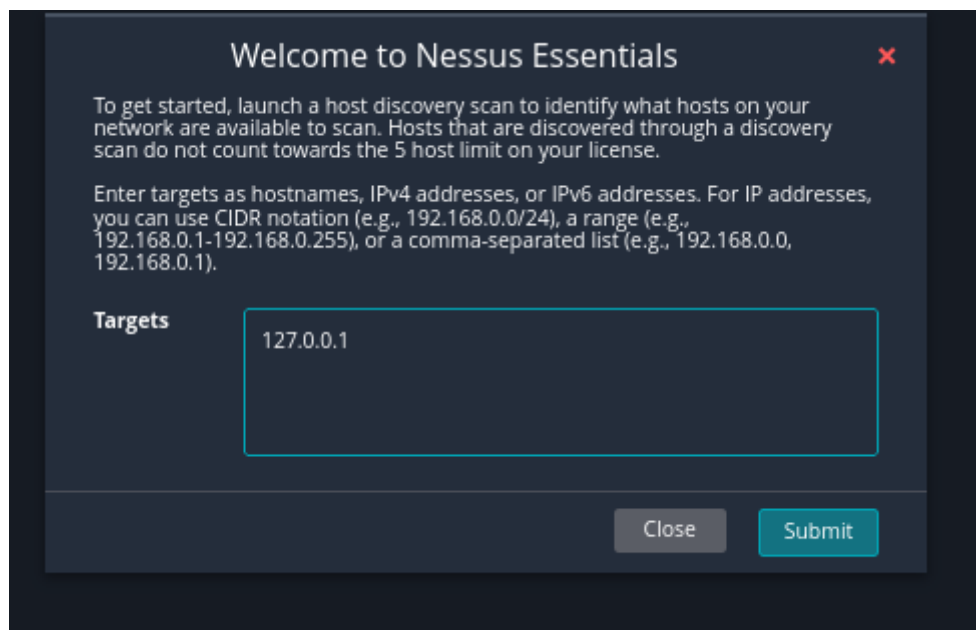
From my understanding, defining scope is very important before performing vulnerability scanning because scanning systems without permission is not allowed and may create legal issues.

For this task, I selected my local Ubuntu system as the target for vulnerability assessment.

Target Details:

- Target IP: 127.0.0.1 (Localhost)
- System: Ubuntu Linux
- Scan Type: Basic Network Scan

Screenshot 1 : Target Configuration Screen



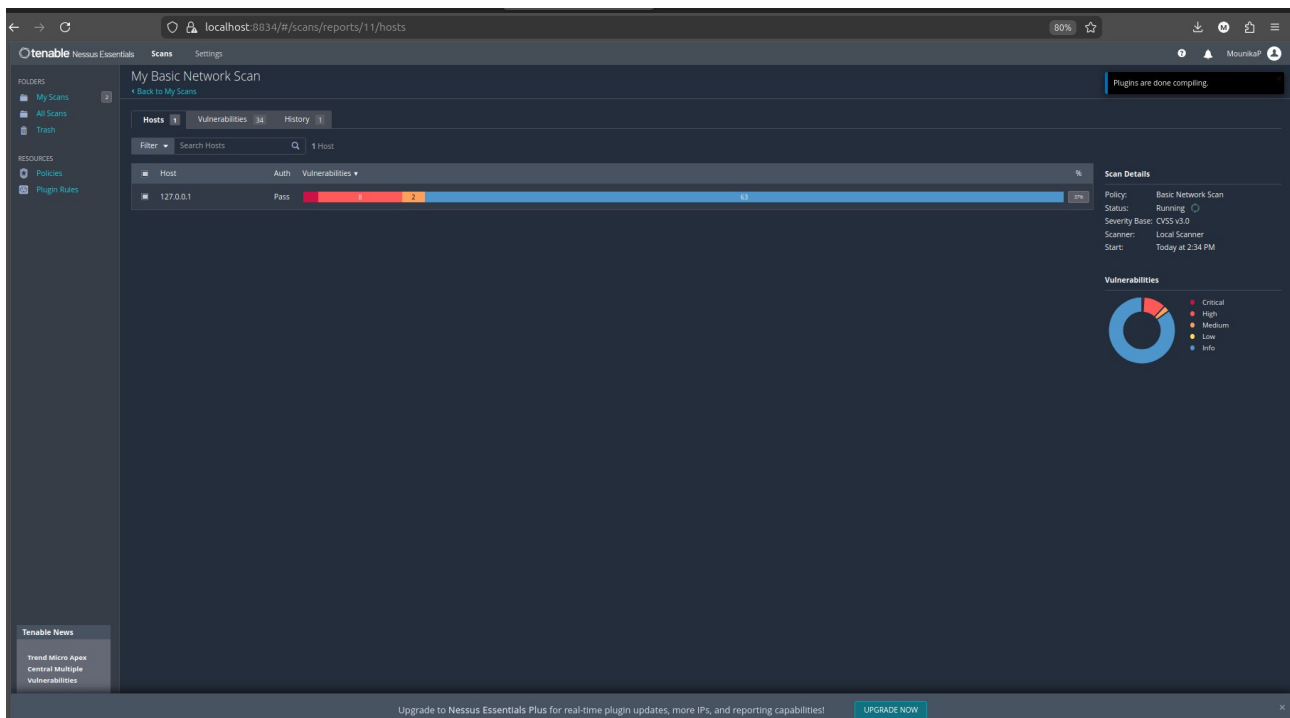
2. Configure Vulnerability Scanner

From my understanding, configuring the scanner correctly ensures accurate vulnerability detection.

I configured Nessus Essentials using:

- Basic Network Scan template
- Default scan settings
- Local scanner option

Screenshot 2 — Scan Configuration / Scan Progress



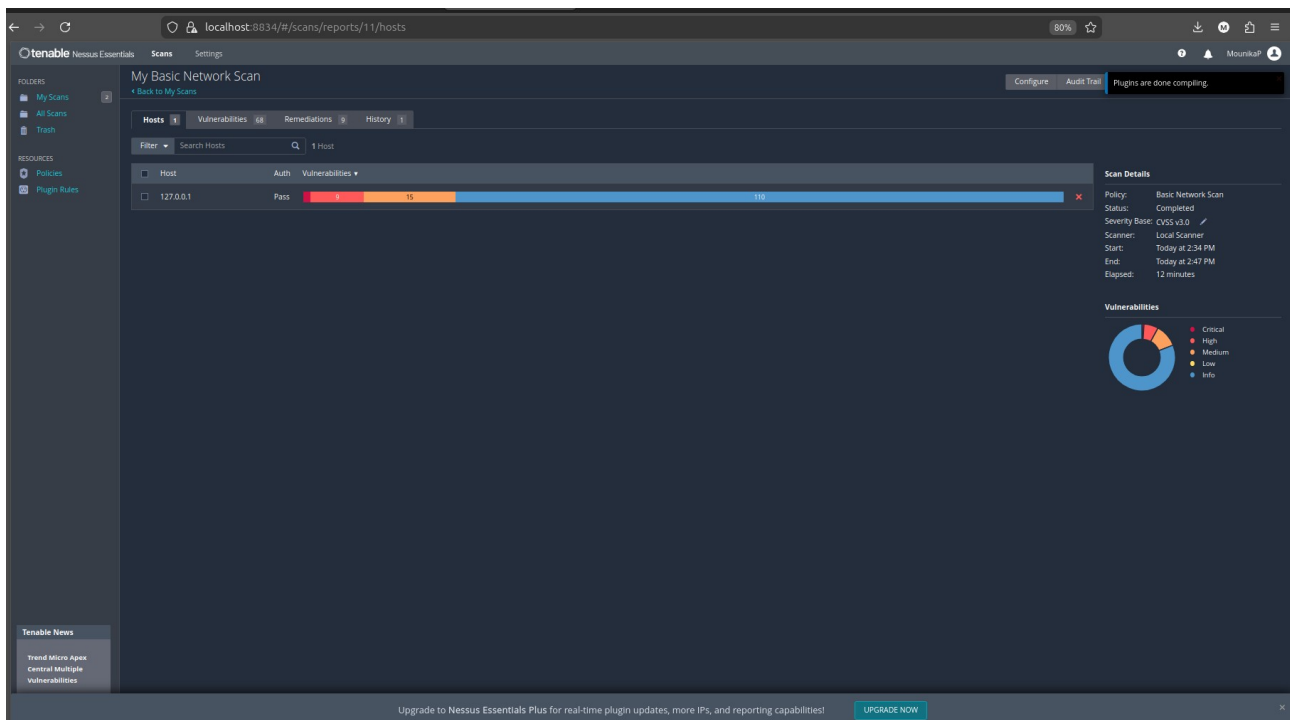
3. Run Vulnerability Scan and Monitor Process

After configuration, I started the scan. During scanning, Nessus performed:

- Host discovery
- Port scanning
- Service detection
- Vulnerability detection

The scan progress showed increasing vulnerability count and percentage completion.

Screenshot 3 — Scan Running Progress:



4. Review Identified Vulnerabilities and Severity Ratings

After the scan was completed, I reviewed vulnerabilities based on severity levels. From my understanding, severity levels help in understanding how dangerous a vulnerability is.

Severity Levels Found:

- Critical
- High
- Medium
- Low
- Informational

Screenshot — Full Vulnerability List

The screenshot displays the Tenable Nessus Essentials interface. The main panel shows a list of vulnerabilities for the host 127.0.0.1. The table includes columns for Severity, CVSS, VPR, EPSS, Name, Family, and Count. The vulnerabilities are categorized by severity: Mixed, High, and Info. The right sidebar provides host details and a vulnerability distribution chart.

Sev	CVSS	VPR	EPSS	Name	Family	Count
Mixed	...	...	...	Canonical Ubuntu Linux (Multiple Issues)	Ubuntu Local Security Checks	10
High	7.5	3.6	0.0004	Python Library Brotli <= 1.1.0 DoS	Misc.	1
Mixed	...	...	...	Apache Log4j (Multiple Issues)	Misc.	13
Mixed	...	...	...	Apache HTTP Server (Multiple Issues)	Web Servers	4
Mixed	...	...	...	SSL (Multiple Issues)	General	4
Info	...	...	...	SSH (Multiple Issues)	General	9
Info	...	...	...	HTTP (Multiple Issues)	Web Servers	5
Info	...	...	...	DMI (Multiple Issues)	General	3
Info	...	...	...	SSH (Multiple Issues)	Misc.	3
Info	...	...	...	TLS (Multiple Issues)	General	2
Info	...	...	...	TLS (Multiple Issues)	Service detection	2
Info	...	...	...	Remote listeners enumeration (Linux / AIX)	Service detection	11
Info	...	...	...	Netstat Portscanner (SSH)	Port scanners	8
Info	...	...	...	OpenJDK Java Detection (Linux / Unix)	General	3

Host Details:
IP: 127.0.0.1
MAC: 28:6B:35:31:8F:C6
OS: Linux Kernel 6.17.0-14-generic on Ubuntu 24.04
Start: Today at 2:34 PM
End: Today at 2:47 PM
Elapsed: 12 minutes
KB: Download
Auth: Pass

Vulnerabilities:
Critical: 1
High: 1
Medium: 0
Low: 0
Info: 15

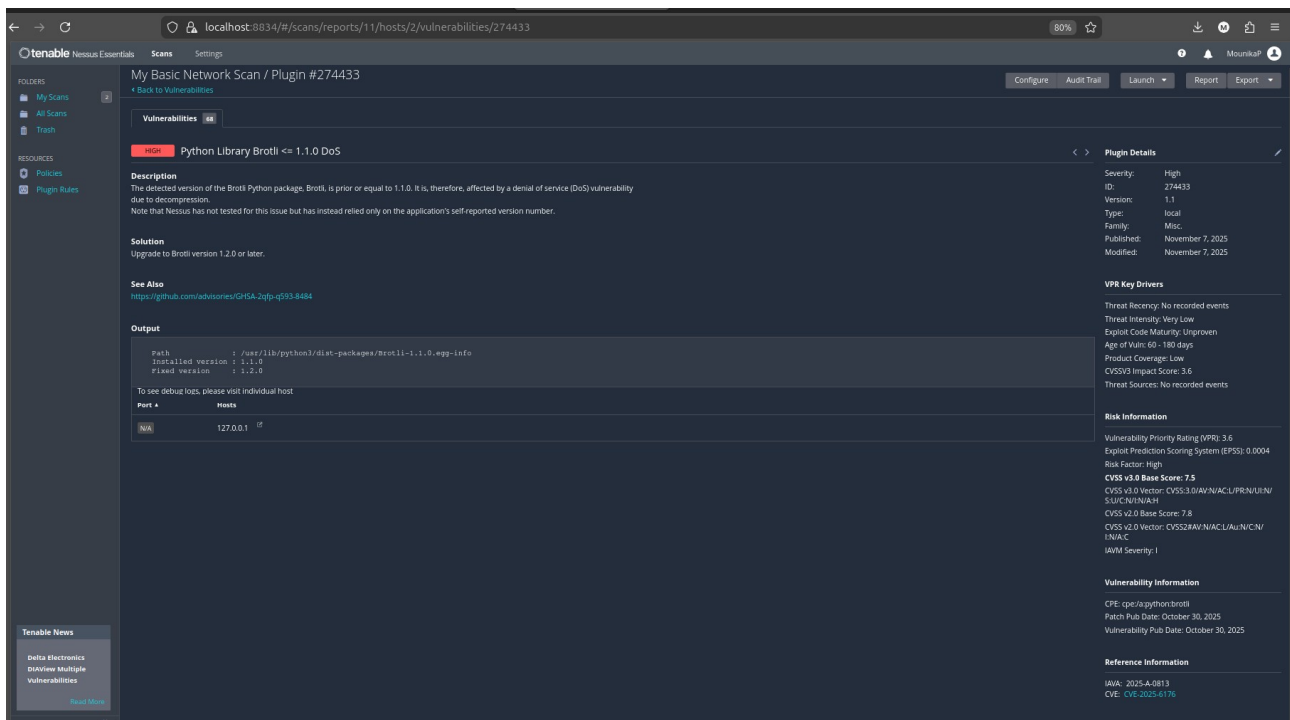
5.Map Findings to CVE and CVSS Scores

From my understanding, CVE gives unique identification to vulnerabilities and CVSS gives severity score from 0–10.

Example Vulnerability Found:

- Python Library Brotli DoS Vulnerability
- CVE ID: CVE-2025-6176
- CVSS Score: High Severity

Screenshot — Vulnerability Detail (CVE + CVSS + Description + Solution)



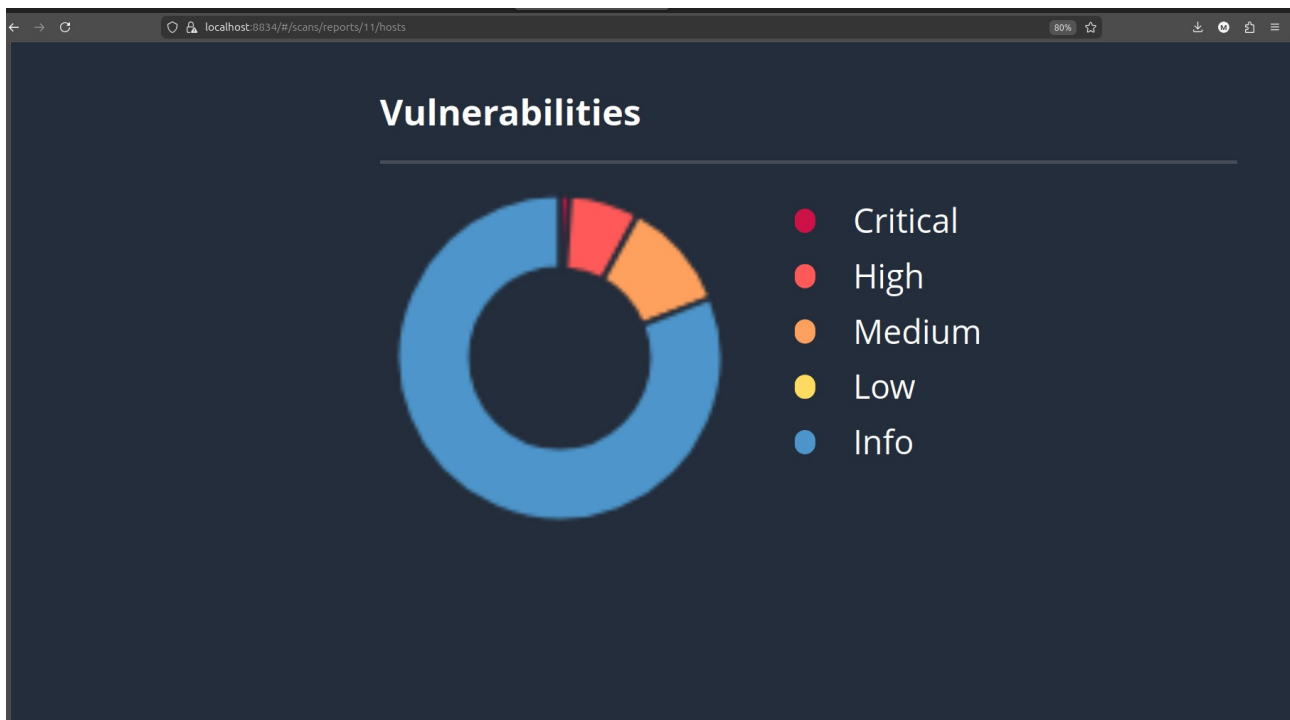
6. Classify Vulnerabilities Based on Risk and Impact

From my analysis, vulnerabilities were classified based on:

- CVSS Score
- Exploit probability
- Risk factor
- System exposure

Severity Distribution was visualized using severity pie chart.

Screenshot — Severity Pie Chart



7. Prioritize Vulnerabilities That Pose Highest Risk

From my understanding, vulnerabilities must be prioritized based on severity and potential impact.

Priority Order Used:

1. Critical vulnerabilities
2. High severity vulnerabilities
3. Medium vulnerabilities
4. Low vulnerabilities

High and Critical vulnerabilities were given highest priority for remediation.

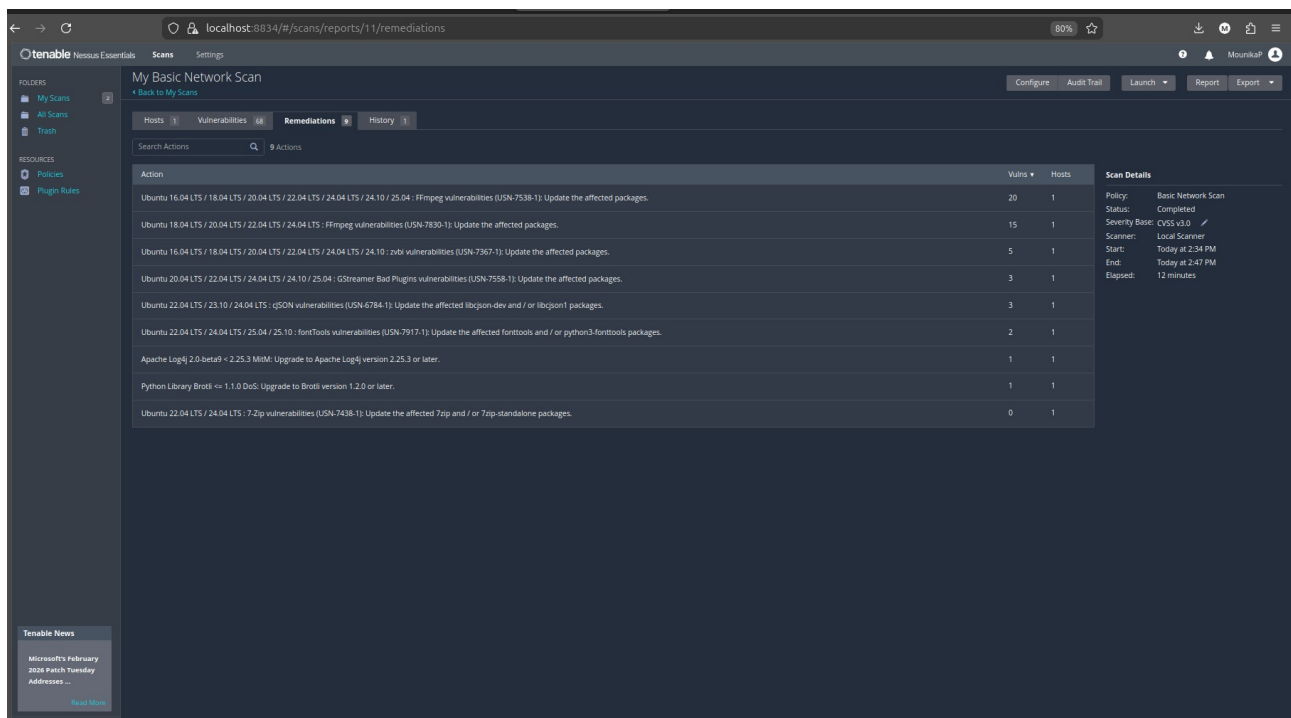
8. Recommend Remediation Steps for Critical Findings

From my understanding, remediation is the most important step because vulnerabilities must be fixed after identification.

Recommended Remediation Steps:

- Update vulnerable software packages
- Apply latest security patches
- Upgrade affected libraries
- Disable unnecessary services
- Enable system firewall

Screenshot — Remediation Tab



9. Risk Priority List

Priority	Vulnerability	Risk Level	Action
1	Python Brotli DoS	High	Update Library
2	Apache Related Issues	Medium	Patch Software
3	SSL Issues	Medium	Update Config
4	Info Issues	Low	Monitor

10. Conclusion

In this task, a vulnerability assessment was performed using Nessus Essentials on an Ubuntu system. Vulnerabilities were identified, analyzed using CVE and CVSS scores, and prioritized based on risk level. Remediation steps were reviewed to help improve the security of the target system.